

■ 시나리오 별 백업 복구(User Managed)

KEY WORDS

- <시나리오 18> 컨트롤파일, 데이터파일 유실, 백업이후 리두정보 X, 비정상 종료
- <시나리오 19> 데이터베이스 정상종료, redo log file, control file 손상되었을 경우 복구
- <시나리오 20> redo log file, control file 손상되어서 데이터베이스가 비정상적인 종료가 된 경우 복구
- <시나리오 21> 백업한 control file 내용과 현재 datafile 정보가 달라졌을 경우 복구
- <시나리오 22> DB 정상 종료 후에 inactive log file 삭제시 복구
- <시나리오 23> inactive log file 삭제된 후 DB 비정상적인 종료 복구
- <시나리오 24> current redo log file 삭제 복구

<<시나리오 18>> 컨트롤파일, 데이터파일 유실, 백업이후 리두정보 X, 비정상 종료

```
SQL> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#, b.next_v$logfile a,
v$log b where a.group# = b.group# order by 1,2;
```

```
SQL> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a, v$tablespace a.ts# =
b.ts#;
```

```
SQL> SQL> select name dbname, open_mode, log_mode, checkpoint_change# from v$database;
```

DBNAME	OPEN_MODE	LOG_MODE	CHECKPOINT_CHANGE#
ORCL	READ WRITE	NOARCHIVELOG	2392446

```
SQL> create table hr.emp5 tablespace users as select * from hr.employees;
Table created.
```

```
SQL> update hr.emp5 set salary=10000;
107 rows updated.
```

```
SQL> SQL> alter system switch logfile;
System altered.
```

```
SQL> /
System altered.
```

```
SQL> /
System altered.
```

```
SQL> ! ls /u01/app/oracle/oradata/ORCL/*.{ctl,dbf}
/u01/app/oracle/oradata/ORCL/control01.ctl /u01/app/oracle/oradata/ORCL/temp01.dbf
/u01/app/oracle/oradata/ORCL/control02.ctl /u01/app/oracle/oradata/ORCL/undotbs01.dbf
/u01/app/oracle/oradata/ORCL/sysaux01.dbf /u01/app/oracle/oradata/ORCL/users01.dbf
```

```
/u01/app/oracle/oradata/ORCL/system01.dbf
```

```
SQL> ! rm /u01/app/oracle/oradata/ORCL/*.ctl,dbf}
```

```
SQL> ! ls /u01/app/oracle/oradata/ORCL/*.ctl,dbf}
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/*.ctl: No such file or directory
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/*.dbf: No such file or directory
```

```
SQL> alter system checkpoint;
```

```
System altered.
```

<Another Session>

```
SQL> select * from v$database;
```

```
select * from v$database
```

```
*
```

```
ERROR at line 1:
```

```
ORA-00210: cannot open the specified control file
```

```
ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

```
ORA-27041: unable to open file
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
Additional information: 3
```

```
SQL> select count(*) from hr.employees;
```

```
select count(*) from hr.employees
```

```
*
```

```
ERROR at line 1:
```

```
ORA-01116: error in opening database file 3
```

```
ORA-01110: data file 3: '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
```

```
ORA-27041: unable to open file
```

```
Linux-x86_64 Error: 2: No such file or directory
```

```
Additional information: 3
```

■ 장애발생 확인 후 복구작업

```
SQL> shutdown abort
```

```
ORACLE instance shut down.
```

```
SQL> ! cp -v /home/oracle/backup/noarch/*.ctl,dbf /u01/app/oracle/oradata/ORCL/
```

```
'/home/oracle/backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

```
'/home/oracle/backup/noarch/control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'
```

```
'/home/oracle/backup/noarch/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
```

```
'/home/oracle/backup/noarch/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
'/home/oracle/backup/noarch/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
```

```
'/home/oracle/backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
```

```
'/home/oracle/backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

SYS@orcl> startup mount

SQL> recover database using backup controlfile until cancel

ORA-00279: change 2390718 generated at 05/07/2025 15:30:40 needed for thread 1

ORA-00289: suggestion :

/u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_66_1199981728.dbf

ORA-00280: change 2390718 for thread 1 is in sequence #66

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

CANCEL

Media recovery cancelled.

SQL> alter database open resetlogs;

Database altered.

SQL> select * from hr.emp5;

select * from hr.emp5

*

ERROR at line 1:

ORA-00942: table or view does not exist

<<시나리오 19>> 데이터베이스 정상종료, redo log file, control file 손상되었을 경우 복구

```
SQL> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#, b.next_change#
from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

```
SQL> create table hr.emp6(id number) tablespace users;
```

Table created.

```
SQL> > insert into hr.emp6(id) values(1);
```

```
SQL> commit;
```

Commit complete.

```
SQL> shutdown immediate
```

■ 장애 발생

```
SQL> ! ls /u01/app/oracle/oradata/ORCL/*.ctl,log}
```

```
/u01/app/oracle/oradata/ORCL/control01.ctl /u01/app/oracle/oradata/ORCL/redo02.log
```

```
/u01/app/oracle/oradata/ORCL/control02.ctl /u01/app/oracle/oradata/ORCL/redo03.log
```

```
/u01/app/oracle/oradata/ORCL/redo01.log
```

```
SQL> ! rm /u01/app/oracle/oradata/ORCL/*.ctl,log}
```

```
SQL> ! ls /u01/app/oracle/oradata/ORCL/*.ctl,log}
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/*.ctl: No such file or directory
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/*.log: No such file or directory
```

```
SQL> startup
```

ORACLE instance started.

Total System Global Area 1811936368 bytes

Fixed Size 8897648 bytes

Variable Size 436207616 bytes

Database Buffers 1358954496 bytes

Redo Buffers 7876608 bytes

ORA-00205: error in identifying control file, check alert log for more info

```
SQL> shutdown abort
```

ORACLE instance shut down.

■ Cre.sql

STARTUP NOMOUNT

CREATE CONTROLFILE REUSE DATABASE "orcl" RESETLOGS NOARCHIVELOG

MAXLOGFILES 16

MAXLOGMEMBERS 3

MAXDATAFILES 100

```
MAXINSTANCES 8
MAXLOGHISTORY 292
LOGFILE
GROUP 1 '/u01/app/oracle/oradata/ORCL/redo01.log' SIZE 10M BLOCKSIZE 512,
GROUP 2 '/u01/app/oracle/oradata/ORCL/redo02.log' SIZE 10M BLOCKSIZE 512,
GROUP 3 '/u01/app/oracle/oradata/ORCL/redo03.log' SIZE 10M BLOCKSIZE 512
DATAFILE
'/u01/app/oracle/oradata/ORCL/system01.dbf',
'/u01/app/oracle/oradata/ORCL/undotbs01.dbf',
'/u01/app/oracle/oradata/ORCL/sysaux01.dbf',
'/u01/app/oracle/oradata/ORCL/users01.dbf'
CHARACTER SET AL32UTF8
;
```

SQL> @cre

ORACLE instance started.

Total System Global Area	1811936368 bytes
Fixed Size	8897648 bytes
Variable Size	436207616 bytes
Database Buffers	1358954496 bytes
Redo Buffers	7876608 bytes

Control file created.

SQL> ALTER DATABASE OPEN RESETLOGS;

Database altered.

```
SQL> ALTER TABLESPACE TEMP ADD TEMPFILE '/u01/app/oracle/oradata/ORCL/temp01.dbf'
      SIZE 125829120 REUSE AUTOEXTEND OFF;
2
```

Tablespace altered.

--백업합시다.

SQL> SHUT IMMEDIATE

Database closed.

Database dismounted.

ORACLE instance shut down.

```
SQL> !cp -v /u01/app/oracle/oradata/ORCL/* /home/oracle/backup/noarch/
'/u01/app/oracle/oradata/ORCL/bnr.sh' -> '/home/oracle/backup/noarch/bnr.sh'
'/u01/app/oracle/oradata/ORCL/bnr1.sh' -> '/home/oracle/backup/noarch/bnr1.sh'
'/u01/app/oracle/oradata/ORCL/control01.ctl' -> '/home/oracle/backup/noarch/control01.ctl'
'/u01/app/oracle/oradata/ORCL/control02.ctl' -> '/home/oracle/backup/noarch/control02.ctl'
```

'/u01/app/oracle/oradata/ORCL/redo01.log' -> '/home/oracle/backup/noarch/redo01.log'
'/u01/app/oracle/oradata/ORCL/redo02.log' -> '/home/oracle/backup/noarch/redo02.log'
'/u01/app/oracle/oradata/ORCL/redo03.log' -> '/home/oracle/backup/noarch/redo03.log'
'/u01/app/oracle/oradata/ORCL/something.lst' -> '/home/oracle/backup/noarch/something.lst'
'/u01/app/oracle/oradata/ORCL/sysaux01.dbf' -> '/home/oracle/backup/noarch/sysaux01.dbf'
'/u01/app/oracle/oradata/ORCL/system01.dbf' -> '/home/oracle/backup/noarch/system01.dbf'
'/u01/app/oracle/oradata/ORCL/temp01.dbf' -> '/home/oracle/backup/noarch/temp01.dbf'
'/u01/app/oracle/oradata/ORCL/undotbs01.dbf' -> '/home/oracle/backup/noarch/undotbs01.dbf'
'/u01/app/oracle/oradata/ORCL/users01.dbf' -> '/home/oracle/backup/noarch/users01.dbf'

<<시나리오 20>> redo log file, control file 손상되어서 데이터베이스가 비정상적인 종료가 된 경우 복구

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#		
	1	19	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT
NO	2326371	1.8447E+19			
	2	17	/u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE
NO	2326365	2326368			
	3	18	/u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
NO	2326368	2326371			

```
SYS@orcl> create table hr.emp7 tablespace users as select * from hr.employees;
```

Table created.

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/
control01.ctl initorcl.ora redo02.log sysaux01.dbf temp01.dbf users01.dbf
control02.ctl redo01.log redo03.log system01.dbf undotbs01.dbf
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/*.{ctl,log}
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/
initorcl.ora sysaux01.dbf system01.dbf temp01.dbf undotbs01.dbf users01.dbf
```

```
SYS@orcl> alter system switch logfile;
```

System altered.

- Alert log file 확인하기

<Another Session>

```
SYS@orcl> select * from v$database;
select * from v$database
```

*

ERROR at line 1:

ORA-00210: cannot open the specified control file

ORA-00202: control file: '/u01/app/oracle/oradata/ORCL/control01.ctl'

ORA-27041: unable to open file

Linux-x86_64 Error: 2: No such file or directory

Additional information: 3

< 원래 Session>

```
SYS@orcl> conn / as sysdba
```

```
SYS@orcl> ! cp -v backup/noarch/*.ctl /u01/app/oracle/oradata/ORCL/
'backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
'backup/noarch/control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'
'backup/noarch/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
'backup/noarch/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
'backup/noarch/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
'backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
'backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
SYS@orcl> shut abort
```

ORACLE instance shut down.

```
SYS@orcl> startup mount
```

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

Database mounted.

```
SYS@orcl> recover database using backup controlfile until cancel;
```

ORA-00279: change 2326564 generated at 05/08/2025 01:00:31 needed for thread 1

ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_19_1200525254.dbf

ORA-00280: change 2326564 for thread 1 is in sequence #19

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

cancel

Media recovery cancelled.

```
SYS@orcl> alter database open resetlogs;
```

Database altered.

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#		
	1	1	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT
NO	2326565	1.8447E+19			
	2	0	/u01/app/oracle/oradata/ORCL/redo02.log	10	UNUSED
YES	0	0			
	3	0	/u01/app/oracle/oradata/ORCL/redo03.log	10	UNUSED
YES	0	0			


```
SYS@orcl> shut abort
```

ORACLE instance shut down.

```
SYS@orcl> ! cp -v backup/noarch/*.{ctl,dbf,log} /u01/app/oracle/oradata/ORCL/  
'backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'  
'backup/noarch/control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'  
'backup/noarch/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'  
'backup/noarch/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'  
'backup/noarch/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'  
'backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'  
'backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'  
'backup/noarch/redo01.log' -> '/u01/app/oracle/oradata/ORCL/redo01.log'  
'backup/noarch/redo02.log' -> '/u01/app/oracle/oradata/ORCL/redo02.log'  
'backup/noarch/redo03.log' -> '/u01/app/oracle/oradata/ORCL/redo03.log'
```

<<시나리오 21>> 백업한 control file 내용과 현재 datafile 정보가 달라졌을 경우 복구

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#		
	1	19	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT
NO	2326371	1.8447E+19			
	2	17	/u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE
NO	2326365	2326368			
	3	18	/u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
NO	2326368	2326371			

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a, v$tablespace b
where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2326567	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2326567	SYSTEM
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2326567	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2326567	ONLINE

```
SYS@orcl> create tablespace sample
2 datafile '/u01/app/oracle/oradata/ORCL/sample01.dbf' size 50m;
```

Tablespace created.

- Alert log file 확인

```
SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a, v$tablespace b
where a.ts# = b.ts#;
```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
3	SYSAUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2326567	ONLINE
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2326567	SYSTEM
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2326567	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2326567	ONLINE
5	SAMPLE	/u01/app/oracle/oradata/ORCL/sample01.dbf	2327716	ONLINE

```
SYS@orcl> shut immediate
```

■ 장애 발생

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/control01.ctl
/u01/app/oracle/oradata/ORCL/control01.ctl
```

```
SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/control01.ctl
```

```
SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/control01.ctl
```

```
ls: cannot access /u01/app/oracle/oradata/ORCL/control01.ctl: No such file or directory
```

```
SYS@orcl> startup
```

```
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
```

```
Fixed Size 8895728 bytes
```

```
Variable Size 754974720 bytes
```

```
Database Buffers 369098752 bytes
```

```
Redo Buffers 7880704 bytes
```

```
ORA-00205: error in identifying control file, check alert log for more info
```

- Alert log file 확인

```
SYS@orcl> shutdown abort
```

■ Recovery

```
SYS@orcl> ! cp -v backup/noarch/*.ctl /u01/app/oracle/oradata/ORCL/
```

```
'backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
```

```
'backup/noarch/control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'
```

```
SYS@orcl> startup mount
```

```
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
```

```
Fixed Size 8895728 bytes
```

```
Variable Size 754974720 bytes
```

```
Database Buffers 369098752 bytes
```

```
Redo Buffers 7880704 bytes
```

```
Database mounted.
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status, b.archived, b.first_change#,
b.next_change# from v$logfile a, v$log b where a.group# = b.group# order by 1,2;
```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#		
1	19	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT
NO	2326371	1.8447E+19		

```

      2      17 /u01/app/oracle/oradata/ORCL/redo02.log      10 INACTIVE
NO      2326365      2326368
      3      18 /u01/app/oracle/oradata/ORCL/redo03.log      10 INACTIVE
NO      2326368      2326371

```

```

SYS@orcl> select a.file#, b.name tbs_name, a.name file_name, a.checkpoint_change#, a.status from v$datafile a, v$tablespace b
where a.ts# = b.ts#;

```

FILE#	TBS_NAME	FILE_NAME	CHECKPOINT_CHANGE#	STATUS
1	SYSTEM	/u01/app/oracle/oradata/ORCL/system01.dbf	2326564	SYSTEM
3	SYS_AUX	/u01/app/oracle/oradata/ORCL/sysaux01.dbf	2326564	ONLINE
4	UNDOTBS1	/u01/app/oracle/oradata/ORCL/undotbs01.dbf	2326564	ONLINE
7	USERS	/u01/app/oracle/oradata/ORCL/users01.dbf	2326564	ONLINE

```

SYS@orcl> alter database open;
alter database openalter database open

```

```

*
```

```

ERROR at line 1:

```

```

ORA-01122: database file 1 failed verification check
ORA-01110: data file 1: '/u01/app/oracle/oradata/ORCL/system01.dbf'
ORA-01207: file is more recent than control file - old control file

```

```

SYS@orcl> recover database using backup controlfile;
ORA-00279: change 2326564 generated at 05/08/2025 01:00:01 needed for thread 1
ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_19_1200525254.dbf
ORA-00280: change 2326564 for thread 1 is in sequence #19

```

```

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

```

```

/u01/app/oracle/oradata/ORCL/redo01.log

```

```

ORA-00283: recovery session canceled due to errors
ORA-01244: unnamed datafile(s) added to control file by media recovery
ORA-01110: data file 5: '/u01/app/oracle/oradata/ORCL/sample01.dbf'

```

```

===> 새로 만든 TS

```

```

ORA-01112: media recovery not started

```

```

SYS@orcl> shut abort
ORACLE instance shut down.

```

```

SYS@orcl> STARTUP NOMOUNT

```

```

SYS@orcl> CREATE CONTROLFILE REUSE DATABASE "ORCL" NORESETLOGS NOARCHIVELOG
MAXLOGFILES 16
MAXLOGMEMBERS 3
MAXDATAFILES 100
MAXINSTANCES 8

```

MAXLOGHISTORY 292

LOGFILE

GROUP 1 '/u01/app/oracle/oradata/ORCL/redo01.log' SIZE 10M BLOCKSIZE 512,

GROUP 2 '/u01/app/oracle/oradata/ORCL/redo02.log' SIZE 10M BLOCKSIZE 512,

GROUP 3 '/u01/app/oracle/oradata/ORCL/redo03.log' SIZE 10M BLOCKSIZE 512

-- STANDBY LOGFILE

DATAFILE

'/u01/app/oracle/oradata/ORCL/system01.dbf',

'/u01/app/oracle/oradata/ORCL/sysaux01.dbf',

'/u01/app/oracle/oradata/ORCL/undotbs01.dbf',

'/u01/app/oracle/product/19.3.0/dbhome_1/dbs/UNNAMED00005', --> "/u01/app/oracle/oradata/ORCL/sample01.dbf"

'/u01/app/oracle/oradata/ORCL/users01.dbf'

CHARACTER SET AL32UTF8

;

-- WARNING! The above command will not work due to the MISSING files.

-- Edit the command to have the correct name, or delete the name.

SYS@orcl> ALTER DATABASE OPEN;

SYS@orcl> ALTER TABLESPACE TEMP ADD TEMPFILE '/u01/app/oracle/oradata/ORCL/temp01.dbf' REUSE;

SYS@orcl> drop tablespace sample including contents and datafiles;

Tablespace dropped.

SYS@orcl> !ls /u01/app/oracle/oradata/ORCL/

control01.ctl initorcl.ora redo02.log sysaux01.dbf temp01.dbf users01.dbf

control02.ctl redo01.log redo03.log system01.dbf undotbs01.dbf

SYS@orcl> ! cp -v backup/noarch/*.{ctl,dbf,log} /u01/app/oracle/oradata/ORCL/ --> 원래대로 돌리기

<<시나리오 22>> DB 정상 종료 후에 inactive log file 삭제시 복구

```

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status,
b.archived, b.first_change#, b.next_change#
from v$logfile a, v$log b
where a.group# = b.group#
order by 1; 2 3 4 5

```

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
			ARC FIRST_CHANGE# NEXT_CHANGE#		
	1	19	/u01/app/oracle/oradata/ORCL/redo01.log	10	CURRENT
NO	2326371	1.8447E+19			
	2	17	/u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE
NO	2326365	2326368			
	3	18	/u01/app/oracle/oradata/ORCL/redo03.log	10	INACTIVE
NO	2326368	2326371			

```

SYS@orcl> select name db_name, open_mode, log_mode,checkpoint_change# from v$database;

```

DB_NAME	OPEN_MODE	LOG_MODE	CHECKPOINT_CHANGE#
ORCL	READ WRITE	NOARCHIVELOG	2326567

■ 장애 발생 (INACTIVE한 하나 삭제)

```

SYS@orcl> shutdown immediate

```

Database closed.

Database dismounted.

ORACLE instance shut down.

```

SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/redo02.log

```

```

SYS@orcl> startup

```

ORACLE instance started.

Total System Global Area 1140849904 bytes

Fixed Size 8895728 bytes

Variable Size 754974720 bytes

Database Buffers 369098752 bytes

Redo Buffers 7880704 bytes

Database mounted.

ORA-03113: end-of-file on communication channel

Process ID: 30936

Session ID: 33 Serial number: 26882

- Alert log file 확인하기

```
SYS@orcl> conn / as sysdba
Connected to an idle instance.
SYS@orcl> startup mount
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
Fixed Size                  8895728 bytes
Variable Size               754974720 bytes
Database Buffers           369098752 bytes
Redo Buffers                7880704 bytes
Database mounted.
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status,
b.archived, b.first_change#, b.next_change#
from v$logfile a, v$log b
where a.group# = b.group#
order by 1; 2 3 4 5
```

	GROUP#	SEQUENCE#	MEMBER	MB STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#	
		1	19 /u01/app/oracle/oradata/ORCL/redo01.log	10 INACTIVE
NO		2326371	2330479	
		2	20 /u01/app/oracle/oradata/ORCL/redo02.log	10 CURRENT
NO		2330479	1.8447E+19	
		3	18 /u01/app/oracle/oradata/ORCL/redo03.log	10 INACTIVE
NO		2326368	2326371	

```
SYS@orcl> alter database clear logfile group 2;
```

Database altered.

```
SYS@orcl> alter database open;
```

Database altered.

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status,
b.archived, b.first_change#, b.next_change#
from v$logfile a, v$log b
where a.group# = b.group#
order by 1; 2 3 4 5
```

	GROUP#	SEQUENCE#	MEMBER	MB STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#	
		1	19 /u01/app/oracle/oradata/ORCL/redo01.log	10 INACTIVE

```

NO      2326371      2330479
          2          0 /u01/app/oracle/oradata/ORCL/redo02.log          10 UNUSED
NO      2330479      2331794
          3          21 /u01/app/oracle/oradata/ORCL/redo03.log          10 CURRENT
NO      2331794      1.8447E+19

```

```
SYS@orcl> alter database drop logfile group 2;
```

Database altered.

```

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status,
b.archived, b.first_change#, b.next_change#
from v$logfile a, v$log b
where a.group# = b.group#
      2      3      4      5
SYS@orcl> /

```

	GROUP#	SEQUENCE#	MEMBER	MB STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#	
		3	21 /u01/app/oracle/oradata/ORCL/redo03.log	10 CURRENT
NO		2331794	1.8447E+19	
		1	19 /u01/app/oracle/oradata/ORCL/redo01.log	10 INACTIVE
NO		2326371	2330479	

```

SYS@orcl> alter database add logfile group 4
'/u01/app/oracle/oradata/ORCL/redo04.log' size 20m;
Database altered.

```

```

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status,
b.archived, b.first_change#, b.next_change#
from v$logfile a, v$log b
where a.group# = b.group#
order by 1;  2      3      4      5

```

	GROUP#	SEQUENCE#	MEMBER	MB STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#	
		1	19 /u01/app/oracle/oradata/ORCL/redo01.log	10 INACTIVE
NO		2326371	2330479	
		3	21 /u01/app/oracle/oradata/ORCL/redo03.log	10 CURRENT
NO		2331794	1.8447E+19	
		4	0 /u01/app/oracle/oradata/ORCL/redo04.log	20 UNUSED
YES		0	0	

```
SYS@orcl> !cp -v /home/oracle/backup/noarch/* /u01/app/oracle/oradata/ORCL/ --> 원래대로 돌리기
```


<<시나리오 23>> inactive log file 삭제된 후 DB 비정상적인 종료 복구

```

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status,
b.archived, b.first_change#, b.next_change#
from v$logfile a, v$log b
where a.group# = b.group#
order by 1; 2 3 4 5

```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#		
1	22	/u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE
NO	2336550	2336560		
2	23	/u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE
NO	2336560	2336586		
3	24	/u01/app/oracle/oradata/ORCL/redo03.log	10	CURRENT
NO	2336586	1.8447E+19	NO	2326368 2326371

```

SYS@orcl> select name db_name, open_mode, log_mode, checkpoint_change# from v$database;

```

DB_NAME	OPEN_MODE	LOG_MODE	CHECKPOINT_CHANGE#
ORCL	READ WRITE	NOARCHIVELOG	2326567

■ 장애 발생(Inactive 상태의 redo 삭제)

```

SYS@orcl> ! rm /u01/app/oracle/oradata/ORCL/redo02.log

```

```

SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/redo02.log

```

```

ls: cannot access /u01/app/oracle/oradata/ORCL/redo02.log: No such file or directory

```

- Alert log file 확인하기

```

SYS@orcl> shut abort

```

```

ORACLE instance shut down.

```

```

SYS@orcl> startup

```

```

ORACLE instance started.

```

```

Total System Global Area 1140849904 bytes

```

```

Fixed Size 8895728 bytes

```

```

Variable Size 754974720 bytes

```

```

Database Buffers 369098752 bytes

```

```

Redo Buffers 7880704 bytes

```

```

Database mounted.

```

```

ORA-00313: open failed for members of log group 2 of thread 1

```

```

ORA-00312: online log 2 thread 1: '/u01/app/oracle/oradata/ORCL/redo02.log'

```

```

ORA-27037: unable to obtain file status

```

Linux-x86_64 Error: 2: No such file or directory

Additional information: 7

```
SYS@orcl> select status from v$instance;
```

STATUS

MOUNTED

```
SYS@orcl> alter database drop logfile group 2;
```

```
alter database drop logfile group 2
```

*

ERROR at line 1:

ORA-01623: log 2 is current log for instance orcl (thread 1) - cannot drop

ORA-00312: online log 2 thread 1: '/u01/app/oracle/oradata/ORCL/redo02.log'

```
SYS@orcl> alter database clear logfile group 2;
```

```
alter database clear logfile group 2
```

*

ERROR at line 1:

ORA-01624: log 2 needed for crash recovery of instance orcl (thread 1)

ORA-00312: online log 2 thread 1: '/u01/app/oracle/oradata/ORCL/redo02.log'

- 데이터 파일에 대해서만 restore

```
SYS@orcl> ! cp -v backup/noarch/*.dbf /u01/app/oracle/oradata/ORCL/
```

```
'backup/noarch/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
```

```
'backup/noarch/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
```

```
'backup/noarch/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
```

```
'backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
```

```
'backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'
```

```
SYS@orcl> recover database until cancel;
```

ORA-00279: change 2326564 generated at 05/08/2025 01:00:31 needed for thread 1

ORA-00289: suggestion : /u01/app/oracle/product/19.3.0/dbhome_1/dbs/arch1_19_1200525254.dbf

ORA-00280: change 2326564 for thread 1 is in sequence #19

Specify log: {<RET>=suggested | filename | AUTO | CANCEL}

cancel

Media recovery cancelled.

```
SYS@orcl> alter database open resetlogs;
```

Database altered.

- redo 확인

```

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status,
b.archived, b.first_change#, b.next_change#
from v$logfile a, v$log b
where a.group# = b.group#
order by 1;

```

GROUP#	SEQUENCE#	MEMBER	MB	STATUS
ARC	FIRST_CHANGE#	NEXT_CHANGE#		

	1	1 /u01/app/oracle/oradata/ORCL/redo01.log	10	ACTIVE
NO	2326565	2330479		
	2	2 /u01/app/oracle/oradata/ORCL/redo02.log	10	CURRENT
NO	2330479	1.8447E+19		
	3	0 /u01/app/oracle/oradata/ORCL/redo03.log	10	UNUSED
YES	0	0		

```

SYS@orcl> ! ls /u01/app/oracle/oradata/ORCL/*.log
/u01/app/oracle/oradata/ORCL/redo01.log /u01/app/oracle/oradata/ORCL/redo03.log
/u01/app/oracle/oradata/ORCL/redo02.log /u01/app/oracle/oradata/ORCL/redo04.log
삭제했던 redo 생성

```

```

SYS@orcl> shut abort
ORACLE instance shut down.

```

```

SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/*.*

```

```

SYS@orcl> !cp -v /home/oracle/backup/noarch/* /u01/app/oracle/oradata/ORCL/
'/home/oracle/backup/noarch/control01.ctl' -> '/u01/app/oracle/oradata/ORCL/control01.ctl'
'/home/oracle/backup/noarch/control02.ctl' -> '/u01/app/oracle/oradata/ORCL/control02.ctl'
'/home/oracle/backup/noarch/initiorcl.ora' -> '/u01/app/oracle/oradata/ORCL/initiorcl.ora'
'/home/oracle/backup/noarch/redo01.log' -> '/u01/app/oracle/oradata/ORCL/redo01.log'
'/home/oracle/backup/noarch/redo02.log' -> '/u01/app/oracle/oradata/ORCL/redo02.log'
'/home/oracle/backup/noarch/redo03.log' -> '/u01/app/oracle/oradata/ORCL/redo03.log'
'/home/oracle/backup/noarch/sysaux01.dbf' -> '/u01/app/oracle/oradata/ORCL/sysaux01.dbf'
'/home/oracle/backup/noarch/system01.dbf' -> '/u01/app/oracle/oradata/ORCL/system01.dbf'
'/home/oracle/backup/noarch/temp01.dbf' -> '/u01/app/oracle/oradata/ORCL/temp01.dbf'
'/home/oracle/backup/noarch/undotbs01.dbf' -> '/u01/app/oracle/oradata/ORCL/undotbs01.dbf'
'/home/oracle/backup/noarch/users01.dbf' -> '/u01/app/oracle/oradata/ORCL/users01.dbf'

```

<<시나리오 24>> current redo log file 삭제 복구

```

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status,
b.archived, b.first_change#, b.next_change#
from v$logfile a, v$log b
where a.group# = b.group#
order by 1;  2    3    4    5

```

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#		
	1	19	/u01/app/oracle/oradata/ORCL/redo01.log	10	ACTIVE
NO	2326371	2330614			
	2	20	/u01/app/oracle/oradata/ORCL/redo02.log	10	ACTIVE
NO	2330614	2336421			
	3	21	/u01/app/oracle/oradata/ORCL/redo03.log	10	CURRENT
NO	2336421	1.8447E+19			

■ 장애 발생

- current 그룹 삭제

```

SYS@orcl> !ls /u01/app/oracle/oradata/ORCL/redo03.log
/u01/app/oracle/oradata/ORCL/redo03.log

```

```

SYS@orcl> !rm /u01/app/oracle/oradata/ORCL/redo03.log

```

```

SYS@orcl> !ls /u01/app/oracle/oradata/ORCL/redo03.log
ls: cannot access /u01/app/oracle/oradata/ORCL/redo03.log: No such file or directory

```

```

SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status,
b.archived, b.first_change#, b.next_change#
from v$logfile a, v$log b
where a.group# = b.group#
order by 1;  2    3    4    5

```

	GROUP#	SEQUENCE#	MEMBER	MB	STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#		
	1	19	/u01/app/oracle/oradata/ORCL/redo01.log	10	INACTIVE
NO	2326371	2330614			
	2	20	/u01/app/oracle/oradata/ORCL/redo02.log	10	INACTIVE
NO	2330614	2336421			
	3	21	/u01/app/oracle/oradata/ORCL/redo03.log	10	CURRENT
NO	2336421	1.8447E+19			

```

SYS@orcl> alter database clear logfile group 3;

```

```
alter database clear logfile group 3
```

```
*
```

```
ERROR at line 1:
```

```
ORA-01624: log 3 needed for crash recovery of instance orcl (thread 1)
```

```
ORA-00312: online log 3 thread 1: '/u01/app/oracle/oradata/ORCL/redo03.log'
```

```
SYS@orcl> shutdown immediate
```

```
Database closed.
```

```
Database dismounted.
```

```
ORACLE instance shut down.
```

```
SYS@orcl> startup
```

```
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
```

```
Fixed Size                  8895728 bytes
```

```
Variable Size              754974720 bytes
```

```
Database Buffers          369098752 bytes
```

```
Redo Buffers               7880704 bytes
```

```
Database mounted.
```

```
ORA-03113: end-of-file on communication channel
```

```
Process ID: 4954
```

```
Session ID: 1 Serial number: 38017
```

```
SYS@orcl> conn / as sysdba
```

```
Connected to an idle instance.
```

```
SYS@orcl> startup mount
```

```
ORACLE instance started.
```

```
Total System Global Area 1140849904 bytes
```

```
Fixed Size                  8895728 bytes
```

```
Variable Size              754974720 bytes
```

```
Database Buffers          369098752 bytes
```

```
Redo Buffers               7880704 bytes
```

```
Database mounted.
```

```
SYS@orcl> alter database clear logfile group 3;
```

```
Database altered.
```

```
SYS@orcl> alter database open;
```

```
Database altered.
```

```
SYS@orcl> select a.group#, b.sequence#, a.member, b.bytes/1024/1024 mb, b.status,  
b.archived, b.first_change#, b.next_change#  
from v$logfile a, v$log b
```

where a.group# = b.group#

order by 1; 2 3 4 5

	GROUP#	SEQUENCE#	MEMBER	MB STATUS
	ARC	FIRST_CHANGE#	NEXT_CHANGE#	
		1	22 /u01/app/oracle/oradata/ORCL/redo01.log	10 CURRENT
NO		2338180	1.8447E+19	
		2	20 /u01/app/oracle/oradata/ORCL/redo02.log	10 INACTIVE
NO		2330614	2336421	
		3	0 /u01/app/oracle/oradata/ORCL/redo03.log	10 UNUSED
NO		2336421	2338180	

SYS@orcl>